MIL-P-50981A(MI)
6 July 1973
SUPERSEDING
MIL-P-50981(MU)
3 January 1972

MILITARY SPECIFICATION

PROPELLANT CHARGE ASSEMBLY

FOR ROCKET, PRACTICE, 35 MM

SUBCALIBER, M73

This specification is approved for use by all activities of the Department of Army,

1. SCOPE

1.1 This specification covers the fabrication of parts and the manufacture of propellant and the assembly of one type of propellant charge designed as Charge Assembly, Propellant, For Use in Rocket, Practice, 35MM, subcaliber, M73.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	Ammunition and Special Weapons General Specification for
MIL-P-14737	Propellant, M7
* MIL-M-51103	Magnesium Oxide, Calcined

FSC 1340

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1235	Single or Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

9272551	Rocket Propellant Charge Assembly
* 9278208	Packing, Marking Prop Chg Assembly Rkt 35 MM Subc M73

PUBLICATIONS

Munitions Command

ET-9272551	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. The rocket propellant charge assembly shall comply with all requirements specified on Drawing (Dwg) 9272551 and with all requirements specified in applicable specifications.

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* 3.3 Composition of Propellant. Composition of propellant shall be in accordance with MIL-P-14737(MU) for propellant Type II. Only Grade A potassium perchlorate may be used. Magnesium oxide shall be in accordance with MIL-M-51103, Type I.

* 3.4 Bond Strength of Stud and Propellant Assembly. The propellant grain shall not become separated from the stud when fired dynamically in a rocket assembly of appropriate weight.

3.5 Maximum pressure at 135°F. The maximum (max.) pressure developed by the propellant charge shall not exceed 12,000 pounds per square inch (psi) when fired in the applicable motor parts assembly and closure at 135°F

* 3.6 Action Time. The action time at minus (-) 10°F shall not exceed 12.5 milliseconds when measured between the 500 psi points on the pressure time trace.

* 3.7 Ballistics. The muzzle velocity of a flight rocket weighing 157.5 ± 0.5 grams, when fired from the applicable launcher at 70°F, shall be 470 ± 10 fps.

*

NOTE: For informational purposes, a witness screen using No. 16 ream weight paper or equivalent shall be placed in position just aft of the rear sight for each firing. Any foreign material blown back on these screens will be cause for referral of the lot for engineering evaluation and decision.

3.8 First article inspection. This specification does not make provisions for a first article inspection.

3.9 Workmanship. The parts and assemblies shall be free from chips, dirt, grease, rust and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to

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MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of change thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind this principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

*

4.2 First article inspection. This specification does not make provisions for a first article inspection.

4.3 Inspection provisions.

4.3.1 Lot formation. The term "Inspection Lot" as used throughout this specification is defined as an essentially homogeneous collection of units of product from which representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions

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from submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination. The examination listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100%. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40% for each major defect and an AQL of 0.65% for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100% inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.3.4.

4.3.2.1 Insert (see Dwg. 10048777, covering a detail of Dwg 9272555).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Concentricity of shank with diameter of head.....	Gage	01001
102.	Height of cut-off tip, max.....	Gage	01002
103.	Diameter of head.....	Gage	01003
104.	Angle of head taper.....	Gage	01004
105.	Serrations missing or incomplete.....	Visual	01005

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Minor:	AQL 1.0 percent		
201.	Length to head angle.....	Gage	01006
202.	Evidence of poor workmanship (see 3.9).....	Visual	01007

4.3.2.2 Assembly, plastic jacketed insert (see Dwg 9272555, covering a detail of Dwg 9272552).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Diameter over plastic, min.....	Gage	02001
102.	Bonding surface incomplete.....	Visual	02002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.9).....	Visual	02003

4.3.2.3 Stud and propellant assembly (see Dwg 9272553, covering a detail of Dwg 9272551).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Total length, max.....	Gage	03001
102.	Cracked grain.....	Visual	03002

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103.	Location of hole.....	Gage	03003

Minor: AQL 0.65 Percent

201.	Evidence of poor workmanship (see 3.9)	03004

4.3.3 Testing.

4.3.3.1 First article sample. This specification does not make provisions for a first article sample.

4.3.3.2 Production.

* 4.3.3.2.1 Chemical Composition and Stability of Propellant (see 3.3) - Major Defect - Code No. 03005. A sample of two pounds of propellant shall be selected from each production lot. This shall be used to perform chemical and stability tests per MIL-P-14737 at a Government-approved laboratory.

* 4.3.3.2.2 Ballistics. (See 3.4 and 3.7) Muzzle velocity at +70°F, Major Defect - Code No. 04001 and bond strength of stud and propellant assembly, Major Defect, Code No. 04002. A sample of 50 propellant charges shall be selected from each lot, assembled to service motors and tested as specified in 4.4.1 using test equipment specified in 4.3.4. Thirty test assemblies shall be conditioned to +70°F and 20 test assemblies shall be conditioned to -10°F. The lot of charge assemblies shall be rejected if the average muzzle velocity of the sample fired at 70°F plus or minus the product of 1.5 and the standard deviation fails to conform to the requirements of section 3.7. The standard deviation of the sample shall be calculated using the formula specified in 6.6. The lot shall also be rejected if one or more propellant sticks become separated from the stud in two or more assemblies tested at -10°F (low velocity round). The occurrence of duds attributed to igniter failure shall require that a replacement test be conducted.

4.3.3.2.3 Static Functioning.

* 4.3.3.2.3.1 Maximum pressure at +135°F (see 3.5), Major Defect, Code No. 04003, and action time at +135°F (for information only), Code No. 04004. A sample of 20 propellant charges shall be selected for each lot and tested as specified in 4.4.2 using test equipment specified in 4.3.4. The lot shall be rejected if any individual test result exceeds the max pressure specified. The pressure and action time shall be recorded and this data forwarded to the cognizant procuring agency or as directed by the contracting officer.

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* 4.3.3.2.3.2 Action time at -10°F (see 3.6), Critical Defect, Code No. 04005, and max pressure at -10°F (for information only, Code No. 04006. A sample of 20 propellant charges shall be selected from each lot and tested as specified in 4.4.2, using test equipment specified in 4.3.4 The lot shall be rejected if any individual test result exceeds the maximum burn time specified in section 3.6. The action time and max pressure shall be recorded and this data forwarded to the cognizant procuring agency or as directed by the contracting officer.

4.3.4 Inspection equipment. Equipment tabulation number ET-9272551 identifies the inspection equipment required to perform the examination and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.5 Government rights to documentation. Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test Methods and Procedures.

* 4.4.1 Ballistics. The propellant charges shall be loaded into numbered service motors with igniters obtained from an established standard lot and assembled to test heads. The complete test rocket shall weigh 157.5 ± 0.5 grams. The test rocket shall be conditioned at the temperature specified in 4.3.3.2.2 ± 2°F for six hours minimum and the test shall be completed in not more than two minutes after removal from the conditioning changer. The test launcher shall be mounted in a rigid horizontal position. Instrumentation to measure the velocity between five and fifteen feet from the muzzle shall be capable of measuring the velocity to ± 1.0 fps. Any round fired at -10°F which fails to achieve a velocity within 10 fps of the average velocity for the sample shall be recovered, opened and examined for evidence of propellant loss from the stud. Observations shall be recorded and reported in accordance with contract requirements.

* 4.4.2 Static Functioning. The plate stud and propellant assembly shall be assembled to the applicable test equipment as specified in 4.3.4 and conditioned at the temperature specified in 4.3.3.2.3 ± 2°F for a min of six hours. The test shall be completed within five minutes after removal from the conditioning chamber. The test equipment shall be capable of obtaining two pressure-time traces simultaneously. Averages of the two traces shall be utilized in the final reduction of data. If one instrumentation channel fails, the results obtained with the remaining channel shall be considered absolute and adequate. Igniters shall be obtained from an established standard lot.

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* 5. Packaging, Packing and Marking. Packaging, packing and marking shall be in accordance with Drawing 9278208.

6. NOTES

6.1 Ordering Data. Procurement documents shall specify the following:

a. Title, number and date of this specification

b. Data cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

6.2 Inspection Code Numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment. The contractor shall design inspection equipment as required by the referenced inspection Equipment List (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment

MU - Army design, mandatory for use.

6.3.3 Army designs. Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when any Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications mandatory (designated "MU") and Non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list reference

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mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

* 6.3.5 Submission of contractor designs. All submitted designs shall contain a reference to the applicable five-digit code number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted to the Government in accordance with appropriate contractual requirements. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Government completion date for design review will be based on the date of the final submission of designs.

6.4 Submission of first article sample. This specification does not make provisions for a first article inspection.

* 6.5 Chemical composition. In the event that it is desirable for informational or other purposes to check the chemical composition of the propellant, the following tests are suggested for best results.

6.5.1 Nitroglycerin. The percentage of nitroglycerin may be determined in accordance with MIL-STD-286, Method 208.1.

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6.5.2 Carbon black. The percentage of carbon black may be determined in accordance with MIL-STD-286, Method 309.1

6.5.3 Nitrocellulose. The percentage of nitrocellulose may be determined in accordance with MIL-STD-286, Method 209.6.

6.6 Calculation of sample standard deviation. The following formula or equivalent shall be used in calculating the sample standard deviations:

$$s = \sqrt{\frac{\Sigma (X - \bar{x})^2}{n-1}}$$

* Where: s = Sample standard deviation

X = Numerical value of individual observations

$\bar{x}$ = Average value of sample

n = Sample size

* 6.7 Inspection and test data. Two copies of the inspection and test data for each lot are to be forwarded to the Government in accordance with the appropriate contract.

* 6.8 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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